



Your Finetuned Large Language Model is Already a Powerful Out-of-Distribution Detector

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Takeaway

Your finetuned large language model (LLM) already functions as a powerful out-of-distribution (OOD) detector. After finetuning an LLM, you obtain a new distribution $p_{\theta'}$ while still having access to the pretrained distribution p_{θ} . For any input sentence x , you can easily calculate the likelihood ratio $p_{\theta}(x)/p_{\theta'}(x)$, as an effective criterion for OOD detection.

Currently, it is straightforward to access both a finetuned LLM and its pre-trained version from online platforms such as Huggingface. Calculating our proposed OOD criterion requires only feeding the input through each model once, with no additional computational cost. Implementing the method requires only **three lines** of code: calculate the log likelihood for each model separately, then subtract them to obtain the criterion.

(Unsupervised) OOD Detection

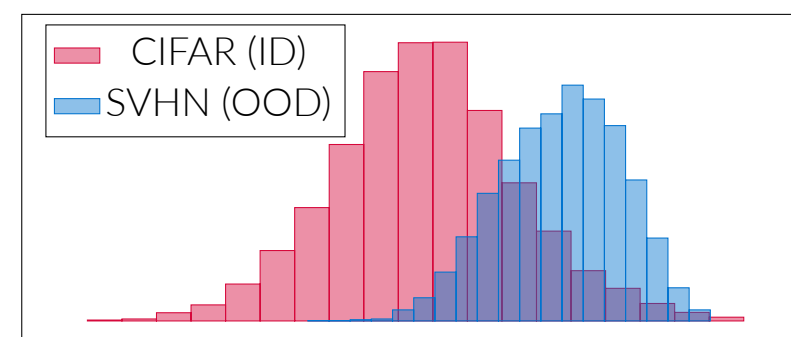
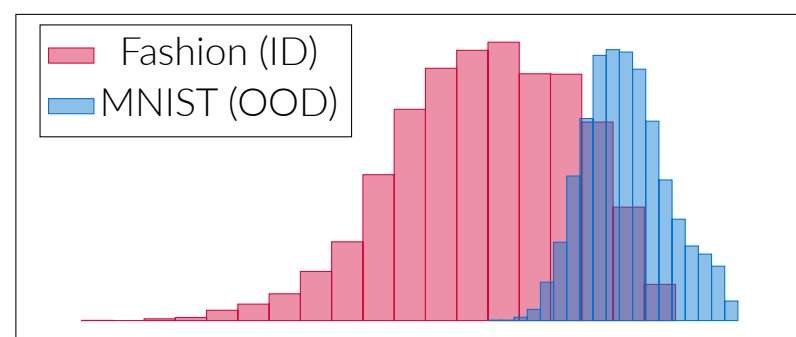
Hendrycks and Gimpel [4] established a baseline for deep learning OOD detection where a model trained on $\mathcal{D}_{\text{in}}^{\text{train}}$ provides a detection criterion S . Performance is evaluated by applying S to samples from $\mathcal{D}_{\text{in}}^{\text{test}} \cup \mathcal{D}_{\text{out}}^{\text{test}}$ and measuring AUROC, AUPR, and FPR95 [11].

The term **unsupervised** refers to the setting where labels for in-distribution data are not accessible.

Nalisnick's Paradox

Given an input x , using the likelihood of in-distribution $p(x)$ as an OOD criterion seems straightforward, since in-distribution data should have higher density within the in-distribution region. In practice, researchers fit a probabilistic generative model (PGM) p_{θ} on $\mathcal{D}_{\text{in}}^{\text{train}}$ and use $p_{\theta}(x)$ as the criterion to detect OOD samples [1].

However, Nalisnick et al. [6] find that for high-dimensional data such as images, sometimes $p_{\theta}(x)$ is higher for samples from $\mathcal{D}_{\text{out}}^{\text{test}}$ than for samples from $\mathcal{D}_{\text{in}}^{\text{test}}$. As illustrated in the figure below, OOD data obtain higher likelihood, making OOD detection ineffective — worse than random guessing.

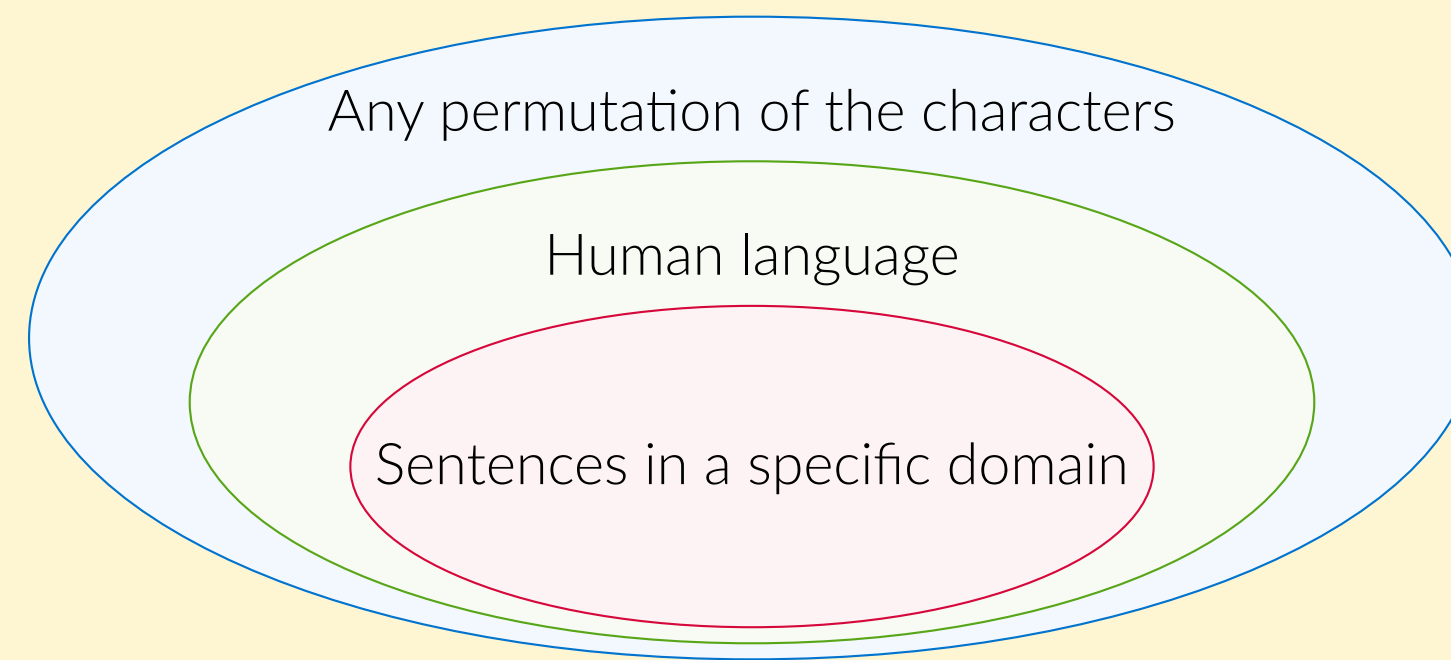


Likelihood Ratio and OOD Proxy

To address this paradox, several studies [8, 10, 9, 13, 2] have proposed using likelihood ratios as the criterion for identifying OOD data. [12] integrates these techniques into a comprehensive structure called the OOD proxy framework. In this framework, we assume that OOD data follow a distribution p_{out} which we cannot directly access. A tractable solution is to build a proxy distribution $p_{\text{out}}^{\text{proxy}}$ to represent p_{out} . The construction of these proxies incorporates empirically-based subjective understanding of OOD data: for example, [8] found that ‘background statistics’ are shared between in-distribution and OOD data, while [13] discovered that local features are shared between in-distribution and OOD data, making these effective OOD proxies. The criterion: $S(x) = p_{\text{out}}^{\text{proxy}}(x)/p_{\text{in}}(x)$.

Pretrained LLM as an OOD Proxy

Given the assumption that pretrained LLMs are trained on a comprehensive corpus of natural languages, we can assert that these models capture the shared features across the entire spectrum of natural languages. This characteristic naturally positions pretrained LLMs as effective OOD proxies.



The figure illustrates how pretrained LLM OOD proxies more accurately distinguish between in-distribution sentences and natural language compared to uniform OOD proxies.

Likelihood Ratio OOD Detection For QA Systems

For QA systems, detecting OOD questions is vital but difficult due to their brevity. Our approach leverages the observation that finetuned LLMs produce reasonable answers for in-distribution questions but unreasonable ones for OOD questions. We propose generating an answer for each question, then applying OOD detection to the question-answer pair rather than the question alone.

Experiments

Due to space limitations, only the results for Near OOD detection are presented here.

Dataset	In-D Label	Model	AUROC $\uparrow$	AUPR $\uparrow$	FPR95 $\downarrow$
ROSTD	No	Gangal et al. [3]	0.981	0.958	0.077
		Jin et al. [5]	0.990	0.973	0.041
		Llama-7B LH	0.960	0.890	0.168
		Llama-7B LR	0.994	0.984	0.023
		Mistral-7B LH	0.964	0.901	0.158
		Mistral-7B LR	0.992	0.978	0.033
		Llama-13B LH	0.965	0.905	0.166
		Llama-13B LR	0.994	0.988	0.018
	Yes	Podolskiy et al. [7]	0.998	0.994	0.008
	SNIPS	No	Gangal et al. [3]	0.955	0.903
Jin et al. [5]			0.963	0.910	0.145
Llama-7B LH			0.912	0.829	0.391
Llama-7B LR			0.993	0.986	0.029
Mistral-7B LH			0.912	0.819	0.417
Mistral-7B LR			0.987	0.968	0.087
Llama-13B LH			0.942	0.872	0.280
Llama-13B LR			0.995	0.988	0.028
Yes		Podolskiy et al. [7]	0.978	0.933	0.120
CLINC150		No	Gangal et al. [3]	0.883	0.677
	Jin et al. [5]		0.902	0.703	0.417
	Llama-7B LH		0.821	0.456	0.538
	Llama-7B LR		0.917	0.766	0.384
	Mistral-7B LH		0.823	0.454	0.540
	Mistral-7B LR		0.913	0.730	0.399
	Llama-13B LH		0.820	0.450	0.546
	Llama-13B LR		0.915	0.742	0.386
	Yes	Podolskiy et al. [7]	0.982	0.939	0.092

From the table, we can observe that using the likelihood ratio between the finetuned model and the pretrained model yields the best performance in the unsupervised OOD detection.

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